Coupled tensor approaches for hyperspectral-multispectral data fusion

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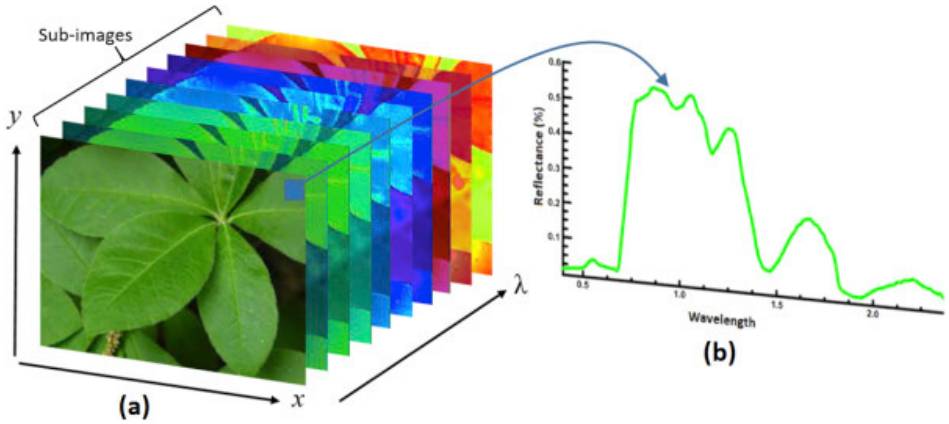
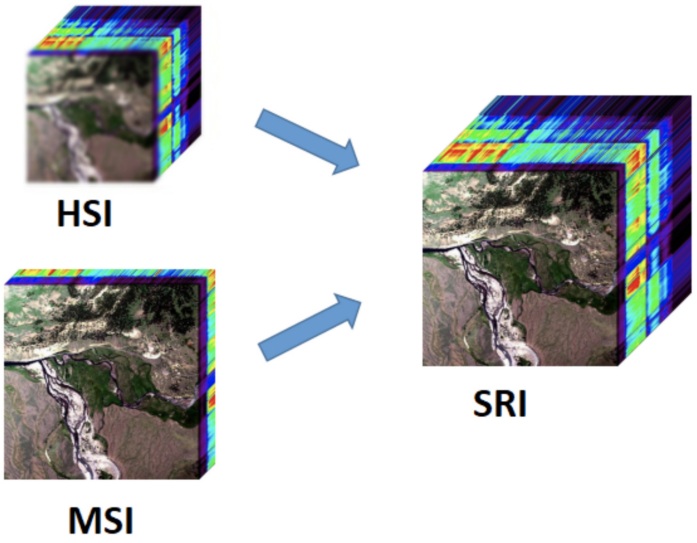
LRMA 19

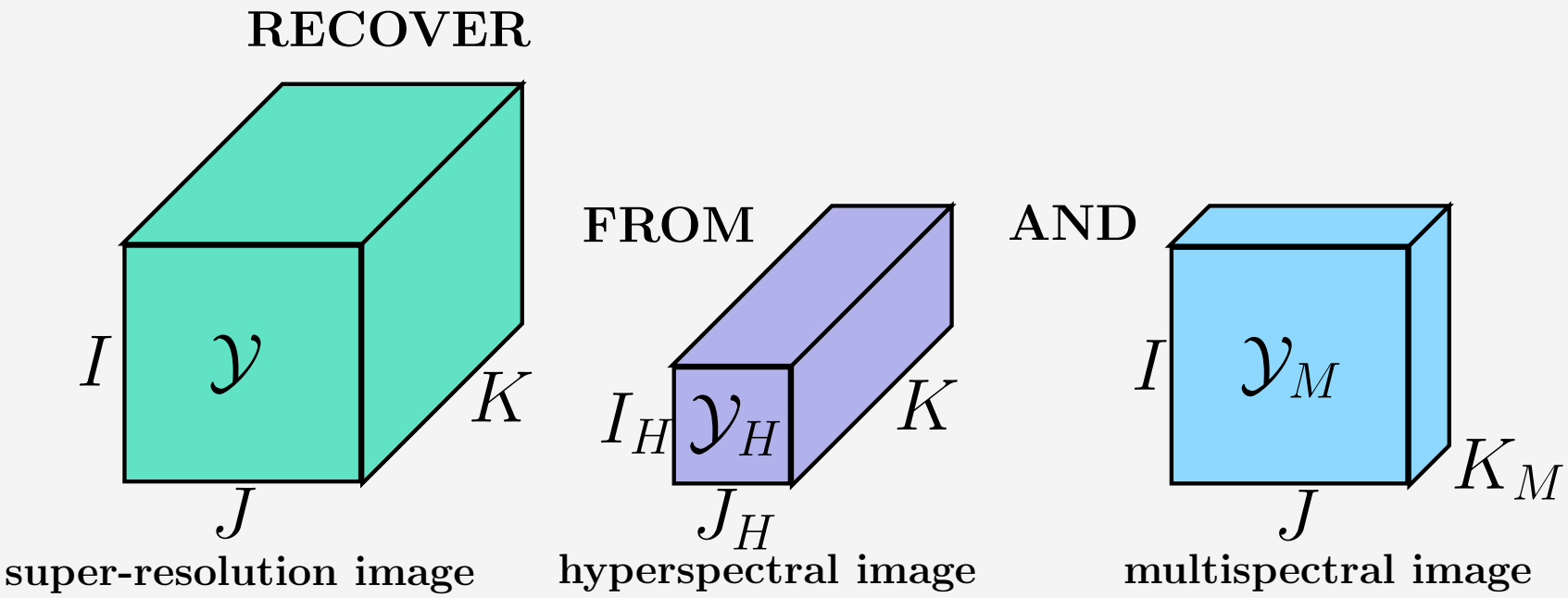
Hyperspectral images





Context

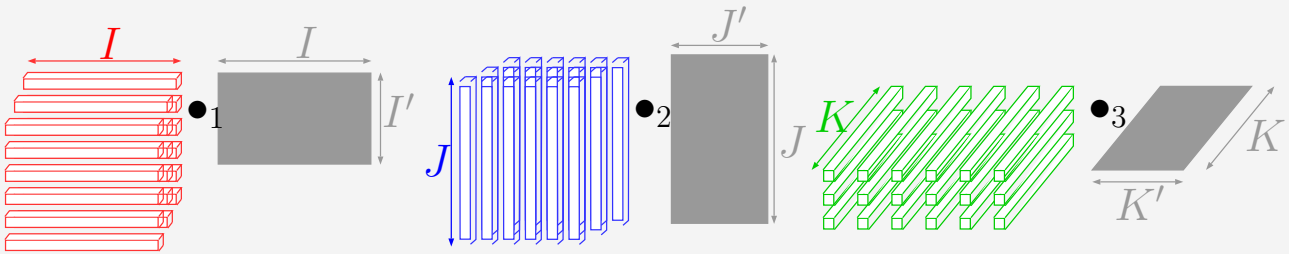
Hyperspectral super-resolution (HSR)



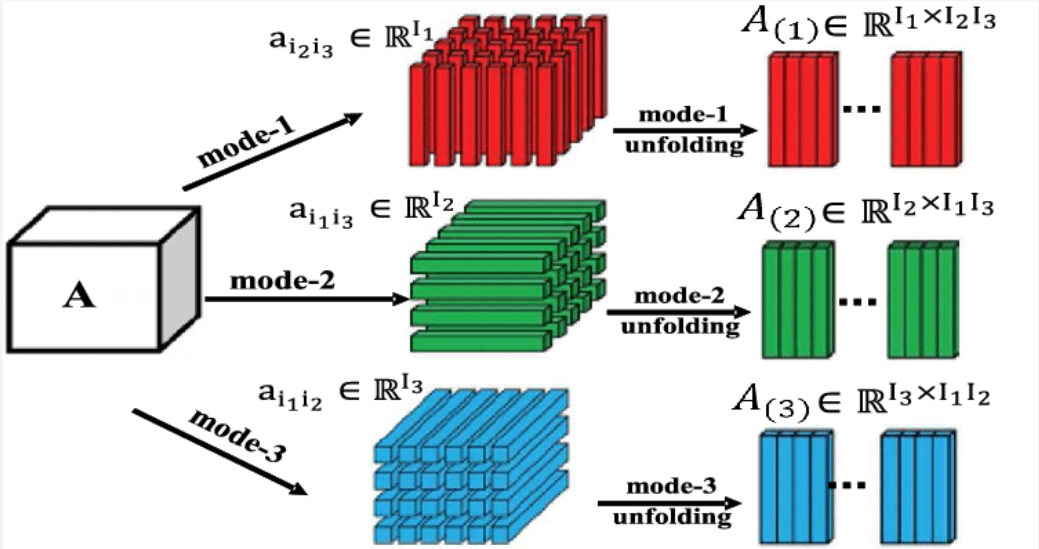
- $\mathcal{Y}_H$: low *spatial* resolution ($I_H J_H < IJ$) ;
- $\mathcal{Y}_M$: low *spectral* resolution ($K_M \ll K$).

Basic tensor operations

Mode product

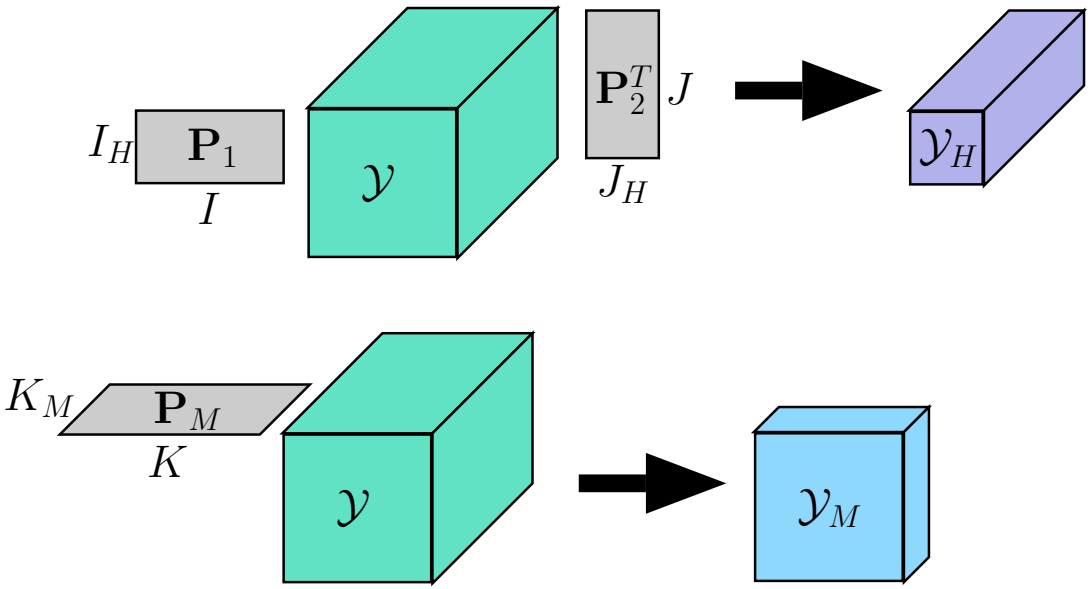


Unfoldings



Degradation model

$$\begin{cases} \mathcal{Y}_H &= \mathcal{Y} \bullet_1 \boldsymbol{P}_1 \bullet_2 \boldsymbol{P}_2 + \mathcal{E}_H, \\ \mathcal{Y}_M &= \mathcal{Y} \bullet_3 \boldsymbol{P}_M + \mathcal{E}_M. \end{cases}$$



- $\boldsymbol{P}_1, \boldsymbol{P}_2$: gaussian blur + downsampling ;
- $\boldsymbol{P}_M$: selection and averaging.

  $\rightarrow$ Fat matrices.

Resolution methods

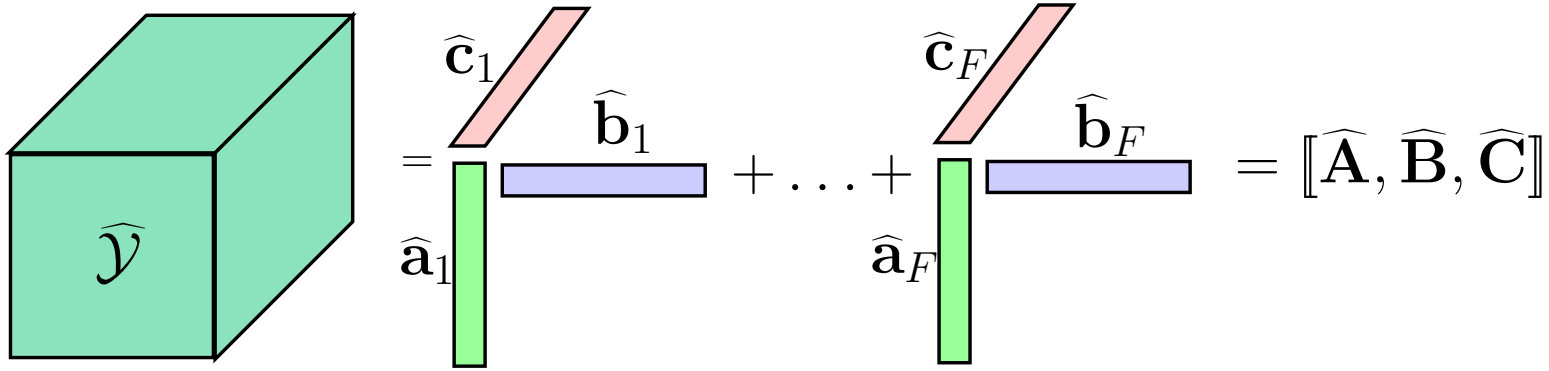
- Matrix approaches : CNMF [Yokoya et al., 2012], HySure [Simoes et al., 2015], FUSE [Wei et al., 2015], FUMI [Wei2016] ;
- Tensor approaches : STEREO [Kanatsoulis et al., 2018], SCUBA [Kanatsoulis et al., 2018](CP decomposition), SCOTT (Tucker decomposition).

Tensor models cost function

$$\underset{\widehat{\mathcal{Y}} \text{ is low rank}}{\text{minimize}} \; f(\widehat{\mathcal{Y}}) = \|\mathcal{Y}_H - \widehat{\mathcal{Y}} \underset{1}{\bullet} \boldsymbol{P}_1 \underset{2}{\bullet} \boldsymbol{P}_2\|_F^2 + \lambda \|\mathcal{Y}_M - \widehat{\mathcal{Y}} \underset{3}{\bullet} \boldsymbol{P}_M\|_F^2.$$

λ : normalized noise level between MSI and HSI.

CP approach



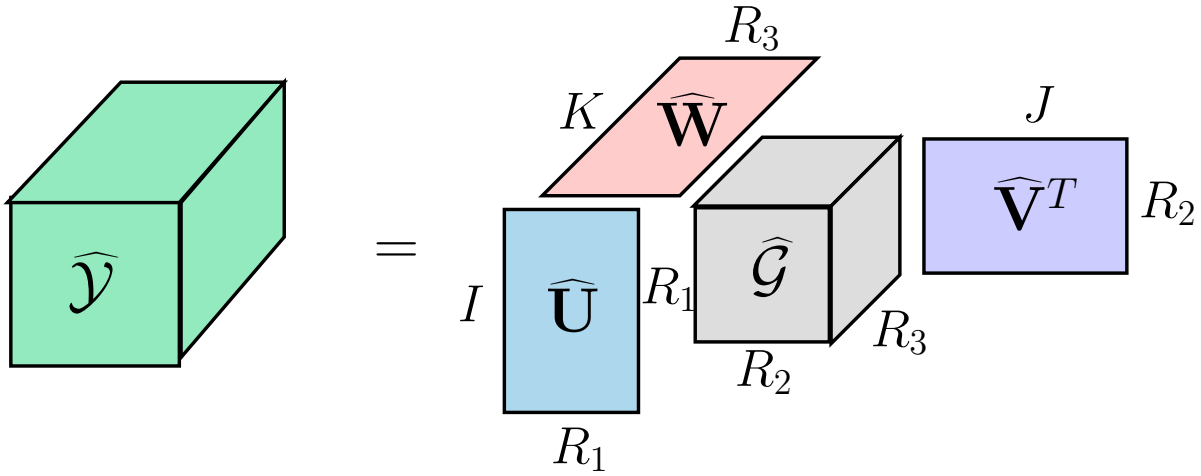
$$\underset{\widehat{\boldsymbol{A}},\widehat{\boldsymbol{B}},\widehat{\boldsymbol{C}}}{\text{minimize}}\ f([\![\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{B}}, \widehat{\boldsymbol{C}}]\!]) = \|\mathcal{Y}_H - [\![\boldsymbol{P}_1\widehat{\boldsymbol{A}}, \boldsymbol{P}_2\widehat{\boldsymbol{B}}, \widehat{\boldsymbol{C}}]\!]\|_F^2 + \lambda\|\mathcal{Y}_M - [\![\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{B}}, \boldsymbol{P}_M\widehat{\boldsymbol{C}}]\!]\|_F^2.$$

STEREO : ALS algorithm on $\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{B}}, \widehat{\boldsymbol{C}}$.

CP approach - drawbacks

- Initialization of STEREO ;
- Iterative algorithm ;
- Non-existence of rank-F decomposition.

Tucker approach



$$= [\![\widehat{\mathcal{G}}; \widehat{\mathbf{U}}, \widehat{\mathbf{V}}, \widehat{\mathbf{W}}]\!]$$

$$\underset{\mathcal{G},\boldsymbol{U},\boldsymbol{V},\boldsymbol{W}}{\text{minimize}}\ f([\![\widehat{\mathcal{G}};\ \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}}]\!]) = \|\mathcal{Y}_H - [\![\widehat{\mathcal{G}};\ \boldsymbol{P_1}\widehat{\boldsymbol{U}}, \boldsymbol{P_2}\widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}}]\!]\|_F^2 + \lambda\|\mathcal{Y}_M - [\![\widehat{\mathcal{G}};\ \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \boldsymbol{P_M}\widehat{\boldsymbol{W}}]\!]\|_F^2.$$

SCOTT : closed-form algorithm

- No iterations ;
- Solution to hyperspectral pansharpening ;
- R_1, R_2, R_3 can be $\neq$.

SCOTT - CLOSED-FORM ALGORITHM

SCOTT

input : $\mathcal{Y}_M$, $\mathcal{Y}_H$, $\boldsymbol{P}_M$, $\boldsymbol{P}_1$, $\boldsymbol{P}_2$; $[R_1, R_2, R_3]$

output: $\widehat{\mathcal{Y}}$

1. $\widehat{\boldsymbol{U}} \leftarrow \text{tSVD}_{R_1}\left(\boldsymbol{Y}_M^{(1)}\right)$, $\widehat{\boldsymbol{V}} \leftarrow \text{tSVD}_{R_2}\left(\boldsymbol{Y}_M^{(2)}\right)$,
   $\widehat{\boldsymbol{W}} \leftarrow \text{tSVD}_{R_3}\left(\boldsymbol{Y}_H^{(3)}\right)$
2. $\widehat{\mathcal{G}} \leftarrow \underset{\mathcal{G}}{\text{argmin}}\ f\left(\llbracket \mathcal{G}; \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}} \rrbracket\right)$
3. $\widehat{\mathcal{Y}} = \llbracket \widehat{\mathcal{G}}; \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}} \rrbracket$

SCOTT – Solving for $\widehat{\mathcal{G}}$

$$\underbrace{\begin{bmatrix} \widehat{\boldsymbol{W}} \boxtimes \boldsymbol{P}_2 \widehat{\boldsymbol{V}} \boxtimes \boldsymbol{P}_1 \widehat{\boldsymbol{U}} \\ \sqrt{\lambda} \boldsymbol{P}_M \widehat{\boldsymbol{W}} \boxtimes \widehat{\boldsymbol{V}} \boxtimes \widehat{\boldsymbol{U}} \end{bmatrix}}_{\boldsymbol{X}} \mathrm{vec}\{\widehat{\mathcal{G}}\} \approx \underbrace{\begin{bmatrix} \mathrm{vec}\{\mathcal{Y}_H\} \\ \sqrt{\lambda}\, \mathrm{vec}\{\mathcal{Y}_M\} \end{bmatrix}}_{\boldsymbol{z}}$$

$$\left(\boldsymbol{X}^\mathsf{T} \boldsymbol{X}\right) \mathrm{vec}\{\widehat{\mathcal{G}}\} = \boldsymbol{X}^\mathsf{T} \boldsymbol{z} \quad (*)$$

with $\boldsymbol{X}^\mathsf{T}\boldsymbol{X} \;=\; \boldsymbol{I}_{R_3} \boxtimes \left(\widehat{\boldsymbol{V}}^\mathsf{T} \boldsymbol{P}_2^\mathsf{T} \boldsymbol{P}_2 \widehat{\boldsymbol{V}}\right) \boxtimes \left(\widehat{\boldsymbol{U}}^\mathsf{T} \boldsymbol{P}_1^\mathsf{T} \boldsymbol{P}_1 \widehat{\boldsymbol{U}}\right) \;+\; \lambda \left(\widehat{\boldsymbol{W}}^\mathsf{T} \boldsymbol{P}_M^\mathsf{T} \boldsymbol{P}_M \widehat{\boldsymbol{W}}\right) \boxtimes \boldsymbol{I}_{R_1 R_2}.$

SCOTT – Solving for $\widehat{\mathcal{G}}$

Generalized Sylvester equation

$$\boldsymbol{A}\widehat{\boldsymbol{G}}\boldsymbol{B} + \boldsymbol{C}\widehat{\boldsymbol{G}}\boldsymbol{D} = \boldsymbol{E}$$

Two ways :

- $\widehat{\boldsymbol{G}} = \boldsymbol{\mathcal{G}}^{(3)} \in \mathbb{R}^{R_1 R_2 \times R_3}$;

$$\boldsymbol{A} = \left(\boldsymbol{U}^{\mathsf{T}}\boldsymbol{P}_1^{\mathsf{T}}\boldsymbol{P}_1\boldsymbol{U}\right) \boxtimes \left(\boldsymbol{V}^{\mathsf{T}}\boldsymbol{P}_2^{\mathsf{T}}\boldsymbol{P}_2\boldsymbol{V}\right), \quad \boldsymbol{B} = \boldsymbol{I}_{R_3},$$
$$\boldsymbol{C} = \boldsymbol{I}_{R_1 R_2}, \quad \boldsymbol{D} = \lambda\left(\boldsymbol{W}^{\mathsf{T}}\boldsymbol{P}_M^{\mathsf{T}}\boldsymbol{P}_M\boldsymbol{W}\right).$$

- $\widehat{\boldsymbol{G}} = \boldsymbol{\mathcal{G}}^{(1)^{\mathsf{T}}} \in \mathbb{R}^{R_1 \times R_2 R_3}$.
  - $\rightarrow$ Equivalent approaches ;
  - $\rightarrow$ Fast solvers [Bartels and Stewart, 1972].

Experimental setup

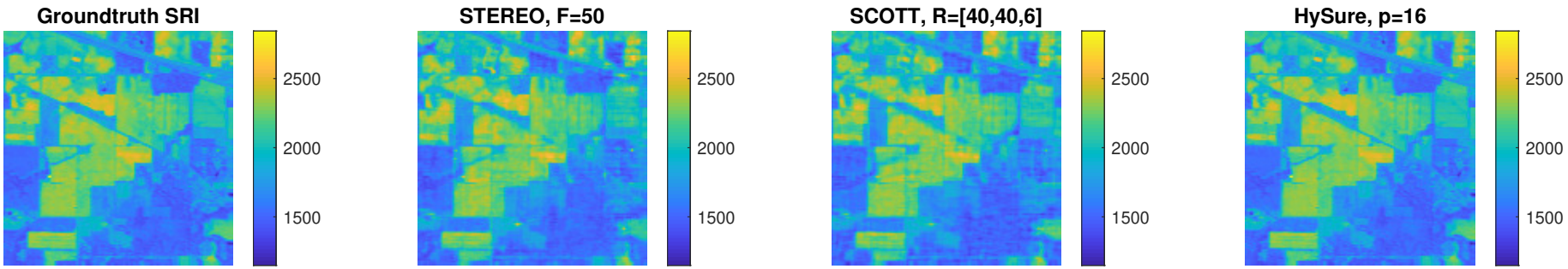
- Groundtruth data : satellite-acquired SRI ;
- Generate $\boldsymbol{P}_1$, $\boldsymbol{P}_2$, $\boldsymbol{P}_M$ such that :
  - $\rightarrow$ $\boldsymbol{P}_M$: selection and averaging of spectral bands (4 to 6 bands) ;
  - $\rightarrow$ $\boldsymbol{P}_1$, $\boldsymbol{P}_2$: gaussian kernel (size $q = 9$), downsampling (ratio $d = 4$).
- Compute $\mathcal{Y}_H$ and $\mathcal{Y}_M$.
- Add noise to $\mathcal{Y}_H$ and $\mathcal{Y}_M$ (optional).

Results - Indian Pines

$\mathcal{Y}_H \in \mathbb{R}^{36\times36\times200}$, $\mathcal{Y}_M \in \mathbb{R}^{144\times144\times6}$

Algorithm	R-SNR	CC	SAM	ERGAS	time (s)
STEREO F=100	28.56	0.92	2.01	0.87	2.1
SCOTT R=[40,40,6]	26.39	0.89	2.32	1.06	0.18
SCOTT R=[30,30,16]	25.15	0.87	2.5	1.18	0.37
SCOTT R=[70,70,6]	27.62	0.9	2.19	0.95	0.33
SCOTT R=[24,24,25]	25.08	0.88	2.44	1.17	0.15
HySure E = 16	28.3	0.92	2.09	0.87	16.06

$$\text{R-SNR} = 10\log_{10}\left(\frac{\|\mathcal{Y}\|_F^2}{\|\hat{\mathcal{Y}} - \mathcal{Y}\|_F^2}\right)$$



Deterministic conditions

$\mathcal{Y}_M = \mathcal{Y} \bullet_3 \boldsymbol{P}_M$ and $\mathcal{Y}_H = \mathcal{Y} \bullet_1 \boldsymbol{P}_1 \bullet_2 \boldsymbol{P}_2$, $\quad \mathcal{E}_M, \mathcal{E}_H = \mathbf{0}$.

Theorem

Let $\mathcal{Y} = [\![\mathcal{G};\ \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{W}]\!]$ be of multilinear rank $[R_1, R_2, R_3]$.

1. If $\text{rank}\{\boldsymbol{Y}_M^{(1)}\} = R_1$, $\text{rank}\{\boldsymbol{Y}_M^{(2)}\} = R_2$, $\text{rank}\{\boldsymbol{Y}_H^{(3)}\} = R_3$ and :
   either a) $\text{rank}\{\boldsymbol{P}_1 \boldsymbol{U}\} = R_1$ and $\text{rank}\{\boldsymbol{P}_2 \boldsymbol{V}\} = R_2$;
   or b) $\text{rank}\{\boldsymbol{P}_M \boldsymbol{W}\} = R_3$.

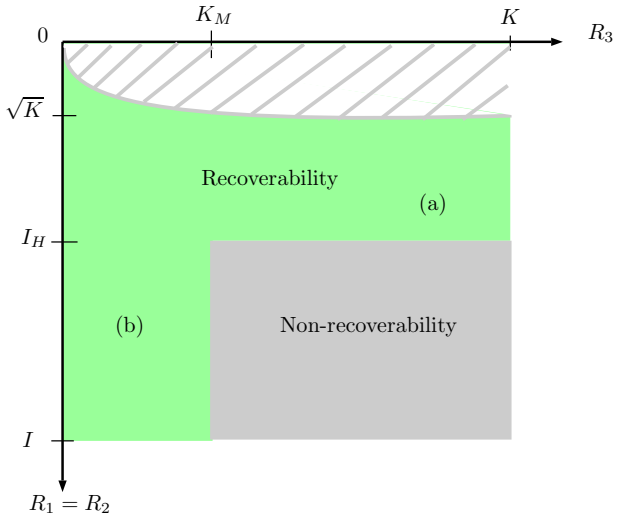
   Then there exists an unique $\widehat{\mathcal{Y}}$ of m.l. rank $[R_1, R_2, R_3]$ such that $\widehat{\mathcal{Y}} \bullet_3 \boldsymbol{P}_M = \mathcal{Y}_M$ and $\widehat{\mathcal{Y}} \bullet_1 \boldsymbol{P}_1 \bullet_2 \boldsymbol{P}_2 = \mathcal{Y}_H$.
2. If neither a) nor b) are satisfied, then $\widehat{\mathcal{Y}}$ is not unique.

Proof : based on determinacy of $\left(\boldsymbol{X}^\mathsf{T} \boldsymbol{X}\right) \text{vec}\{\widehat{\mathcal{G}}\} = \boldsymbol{X}^\mathsf{T} \boldsymbol{z}$.

Generic conditions with $R_1 = R_2,\ I_H = J_H,\ I = J$

We suppose that

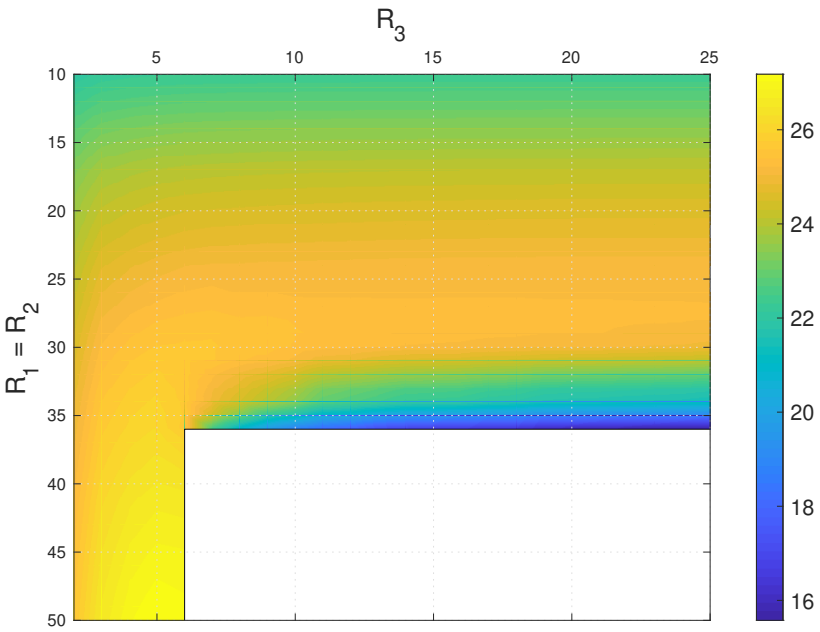
- $\boldsymbol{P}_1, \boldsymbol{P}_2, \boldsymbol{P}_M$ are full-rank matrices, $\boldsymbol{\mathcal{E}}_M, \boldsymbol{\mathcal{E}}_H = \boldsymbol{0}$;
- $\boldsymbol{\mathcal{G}}$, $\boldsymbol{U}$, $\boldsymbol{V}$ et $\boldsymbol{W}$ are random ;



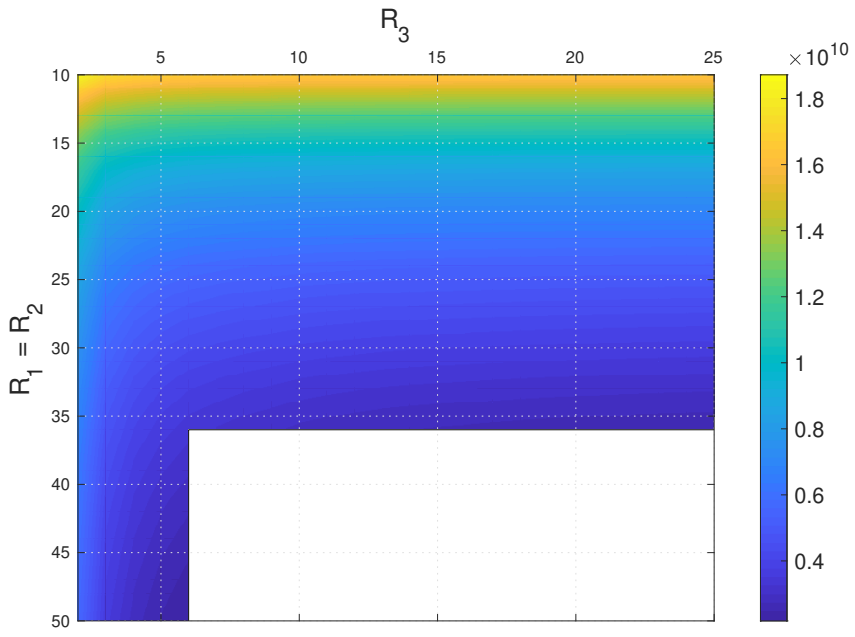
$$\left.\begin{array}{l} R_3 \leq \min(R_1, I_H)^2, \\ a) R_1 \leq I_H \text{ or } b) R_3 \leq K_M \end{array}\right\} \Rightarrow \widehat{\boldsymbol{\mathcal{Y}}} \text{ is unique}$$

a) and b) not satisfied $\Rightarrow$ $\widehat{\boldsymbol{\mathcal{Y}}}$ is non-unique (*non recoverable*).

Illustration for Indian Pines



R-SNR



$f\left(\widehat{\mathcal{Y}}\right)$

Illustration for Indian Pines

$$\mathrm{cond}(\boldsymbol{X}^\top \boldsymbol{X}) := \frac{\sigma_{max}(\boldsymbol{X}^\top \boldsymbol{X})}{\sigma_{min}(\boldsymbol{X}^\top \boldsymbol{X})}$$

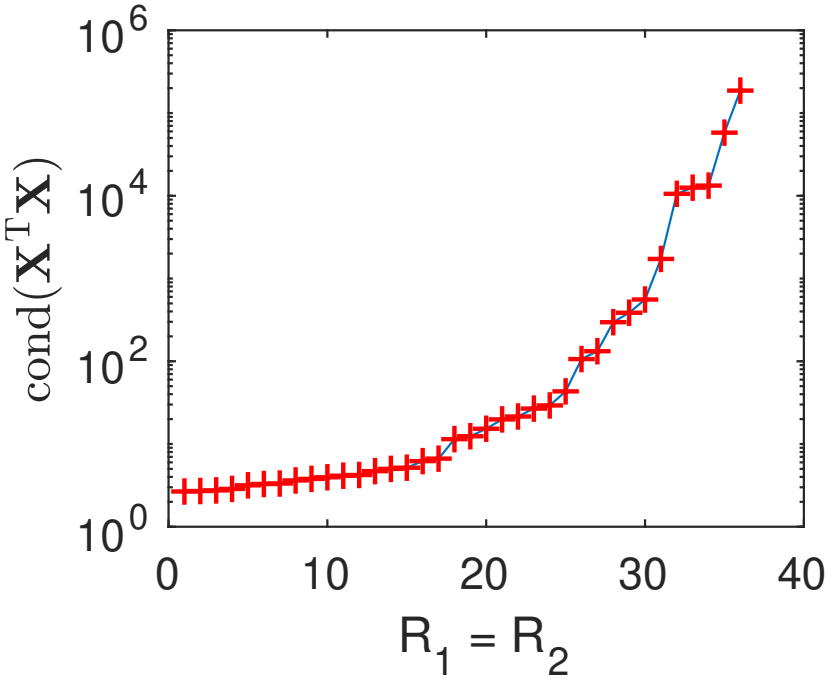


Figure – G : $\mathrm{cond}(\boldsymbol{X}^\top \boldsymbol{X})$ as function of $R_1 = R_2$, fixed $R_3 = K_M$

BSCOTT - "BLIND" ALGORITHM ($\boldsymbol{P}_1$, $\boldsymbol{P}_2$ UNKNOWN)

BSCOTT

input : $\mathcal{Y}_M$, $\mathcal{Y}_H$, $\boldsymbol{P}_M$; $[R_1, R_2, R_3]$

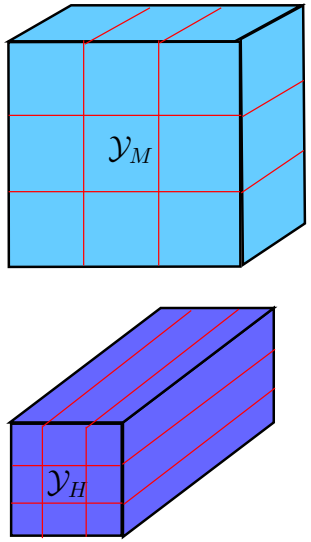
output: $\widehat{\mathcal{Y}}$

Split $\mathcal{Y}_M$, $\mathcal{Y}_H$ into L blocks along spatial dimensions.

for l=1 :L **do**

1. $[\![\widehat{\mathcal{G}}_l; \widehat{\boldsymbol{U}}_l, \widehat{\boldsymbol{V}}_l, \widetilde{\boldsymbol{W}}_l]\!] \overset{\text{HOSVD}}{\approx} \mathcal{Y}_{M,l}$
2. $\boldsymbol{Z}_l \leftarrow \text{tSVD}_{R_3}\left(\boldsymbol{Y}_{H,l}^{(3)}\right)$
3. $\widehat{\boldsymbol{W}}_l \leftarrow \boldsymbol{Z}_l(\boldsymbol{P}_M\boldsymbol{Z}_l)^{\dagger}\widetilde{\boldsymbol{W}}_l$
4. $\widehat{\mathcal{Y}}_l = [\![\widehat{\mathcal{G}}_l; \widehat{\boldsymbol{U}}_l, \widehat{\boldsymbol{V}}_l, \widehat{\boldsymbol{W}}_l]\!]$

end

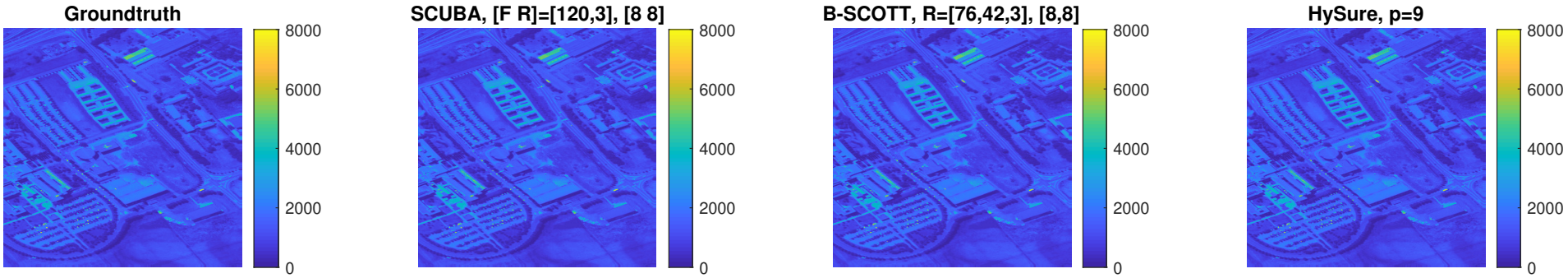


$R_3 \leq K_M$

Results – Pavia University

$\mathcal{Y}_H \in \mathbb{R}^{152\times84\times103}$, $\mathcal{Y}_M \in \mathbb{R}^{608\times366\times4}$

Algorithm	R-SNR	CC	SAM	ERGAS	time (s)
SCUBA (4,4) [F,R3]=[120,3]	25.67	0.99	3.25	1.97	17.6
B-SCOTT (4,4) R=[152,84,3]	26.42	0.99	2.97	1.83	0.55
SCUBA (8,8) [F,R3]=[120,3]	26.49	0.99	2.93	1.81	49.35
B-SCOTT (8,8) R=[76,42,3]	26.52	0.99	2.92	1.84	0.73
Blind-STEREO F = 300	23.14	0.98	3.96	2.45	66.97
HySure E = 9	26.05	0.99	2.9	1.79	101.54



Conclusion

- Similar performances to CP approach ;
- Non-unique choice of m.l. rank ;
- Conditions on unicity $\widehat{\mathcal{Y}}$;
- Solution to hyperspectral pansharpening ;

Linked publications :

Coupled Tensor Low-rank Multilinear Approximation for Hyperspectral Super-resolution - C. Prévost, K. Usevich, P. Comon and D. Brie. ICASSP 2019, Brighton, United Kingdom ;

Approches tensorielles coupleés pour la fusion aveugle d'images multispectrale et hyperspectrale - C. Prévost, K. Usevich, P. Comon and D. Brie. GRETSI 2019, Lille, France

Hyperspectral Super-Resolution with Coupled Tucker Approximation : Identifiability and SVD-based algorithms - C. Prévost, K. Usevich, P. Comon and D. Brie (Preprint).

Thank you for your attention !